

ASSEMBLYAI VOICE AGENT HACKATHON

Tell

It won't take “yes” for an answer.

A medication-adherence follow-up call that listens to **how** the answer was given — and refuses to write it into the record until the patient sounded like they meant it.

THE PROBLEM

Ask a patient if they took their tablets, and they say yes.

Self-reported adherence overstates the real thing by a wide margin, and every clinician who has made a follow-up call knows it.

Which is why a good nurse doesn't listen to the word "yes".

They listen to the half-second before it.

WHAT EVERY VOICE AGENT DOES WITH THAT HALF-SECOND

It throws it away.

Patient “Uh 900 ms yeah, yeah, I’ve been taking them.”

Transcribes
yes

The record ends up holding a fact that nobody actually believes —
and the next clinical decision is made on top of it.

The agent cannot write to the record by itself.

It has to call a tool, and that tool is gated on how the answer sounded.

No adherence fact enters the record unless the patient sounded certain — or was asked again.

A refusal isn't an error message. It's a **specific follow-up question**, chosen by whichever signal fired, handed back to the agent in the `error` field of the tool result.

We control when `tool.result` is sent, so the gate holds the agent's turn open until it has actually heard the patient. **No race, no guessing.**

A REAL RUN · NODE TOOLS/SIMULATE-CALL.MJS · UNEDITED

Patient “Uh 900 ms yeah, yeah, I’ve been taking them.” 1.9 s to start

agent → record_adherence { answer: "took_all" }

GATE · REFUSED

1668 ms dead air – “Ah.” held for 1696 ms
transcription confidence fell to 0.61 on “been”

→ probe: ask which specific days they missed

34%
certain

Tell “Do you know if there were any specific days you might have missed?”

Patient “Okay, honestly – I missed Tuesday and Wednesday.”

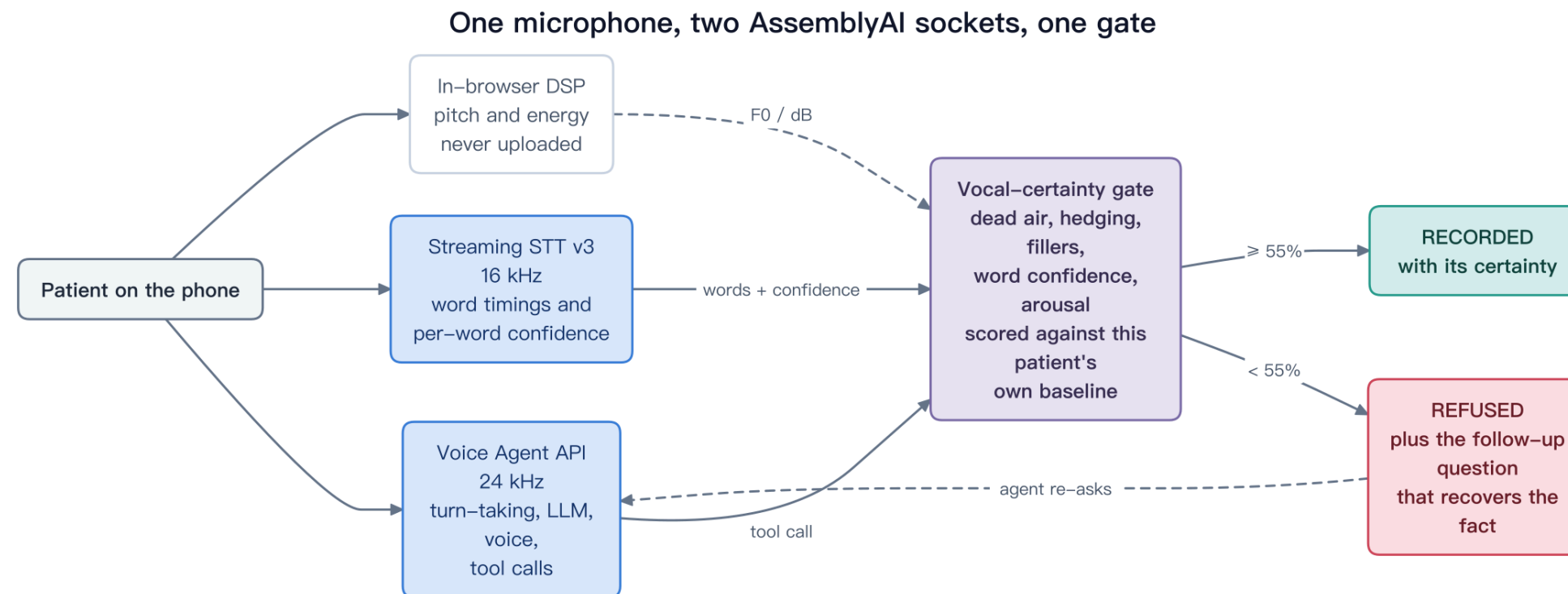
GATE · RECORDED

metformin: missed_some (Tuesday, Wednesday)

88%

One microphone, two AssemblyAI sockets.

The Voice Agent API returns `transcript.user` as a plain string — everything an agent needs to answer a question, and nothing you need to judge whether someone meant it. Streaming v3 returns the same speech as **words with millisecond boundaries and a confidence each**. One socket runs the conversation; the other measures it.



The Voice Agent API owns the conversation. Streaming STT v3 owns the measurement. Neither can write to the record without the other.

Eight signals, combined with a noisy-OR — so one loud tell isn’t averaged away by seven quiet ones.

SIGNAL	WHAT IT CATCHES
onsetDelay	how long before they answered at all
preAnswerPause	dead air — gaps and drawled fillers
hedging	“I think”, “pretty much”, “I try to”
fillerLoad	“uh”, “um”, weighted
wordConfidence	the transcriber’s own uncertainty
revision	started one answer, switched to another
arousal	pitch and loudness off their baseline
brevity	a one-word answer that took a long time

Every signal is scored against **this patient, earlier in this same call.**

A patient who opens every sentence at 2.2 seconds isn’t penalised for taking 2.2 seconds. A patient who’s been answering in 300 ms and suddenly takes 2.2 **is**.

Ordinary turns — the small talk, the consent — are what build that baseline. Before there’s enough of them, scoring falls back to absolute thresholds and is **labelled as weaker evidence**, in the UI and in the record.

FOUR THINGS WE FOUND · NONE OF THEM DOCUMENTED

FINDING

Sentiment analysis is prosody-blind; word confidence isn't.

Eight RAVDESS readings of one sentence, neutral to terrified, all return NEUTRAL within 0.07.

WHY IT MATTERED

min word confidence
0.994 calm → 0.504 fearful
the signal was in the response all along

Under stress the model writes down disfluencies nobody said.

The fearful reading comes back as “Kids uh are talking by the door”.

there is no “uh”
in the RAVDESS script

Streaming v3 rejects audio frames outside 50–1000 ms.

An AudioWorklet naturally emits ~21 ms blocks.

error 3007, socket closed
the call dies the moment they speak

Silence is swallowed into the preceding token.

A 900 ms hesitation comes back as ONE word, "Ah.", spanning 1696 ms — only 128 ms of measurable gap.

86% certain → 34%
once dead air is counted properly

AND ONE WE THREW OUT

We'd been carrying a fifth finding for a week. It didn't hold up.

We believed `language_detection: true` silently disabled disfluency preservation. Before publishing, we went back to the raw responses.

RUN	WORDS	FILLERS
language_detection	401	0
language_code: en_us	412	0

Neither transcript had a single “uh”. The 11-word difference was **one dropped clause**, not stripped disfluencies.

We'd rather ship one fewer headline than one we can't defend.

It's written up in the repo as the thing we got wrong — `docs/evidence.md` §5.

WHERE IT STOPS, AND WHAT IT ISN'T

Designed to be refused

- **One re-ask, then it accepts.** A third time is an interrogation, not care — it records and flags for a clinician instead.
- **The patient is never told this exists.** No mention of pauses, tone or scores. The prompt forbids it.
- **Hesitation is not deception.** A low score means this answer isn't reliable enough to act on — a claim about evidence, not about the person.

Honest limits

- **No clinical validation.** Every number here comes from RAVDESS actors and synthesised speech. No real patient has used it.
- **Accent, age, stammer and cognitive conditions move the signals.** Scoring within one call is a mitigation, not a solution.
- **A flagged answer is a prompt for a human to look** — never a conclusion.

Tell

It won't take "yes" for an answer.

github.com/qiqiiiiii77/tell

39 tests · zero dependencies · MIT